

LAXMAN SAI AVIRESH

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Education

VNR Vignana Jyothi Institute of Engineering and Technology

2022 – 2026

B.Tech in Computer Science & Engineering (AI & ML); CGPA: 8.80/10.0

Hyderabad, India

Sri Chaitanya Junior College

2019 – 2021

Intermediate; Percentage: 94.3%

Hyderabad, India

Work Experience

Algo University (backed by Y-Combinator)

May 2025 – July 2025

SWE Co-Op Intern

Remote

- Developed **CodeJudge**, a full-stack Online Judge platform using the **MERN stack** that supports **50+ users** and a library of **50+ algorithmic challenges**.
- Implemented secure user authentication with **JWT** and designed scalable **RESTful APIs** for problem management and submission evaluation.
- Built a **custom multi-language compiler** (C, C++, Python, Java) without third-party APIs to ensure fast and sandboxed code execution.
- Integrated an **AI-driven code review module** that provides automated feedback on code quality and efficiency, helping users optimize their solutions post-submission.
- Containerized services using **Docker**, and deployed the system on **AWS EC2** with images stored in **AWS ECR** for scalable, cloud-native performance.

Projects

Chat App | *MERN Stack, Socket.io*

- Built a real-time chat application using the **MERN stack** (MongoDB, Express.js, React.js, Node.js) and **Socket.io** to enable secure, bi-directional communication between users.
- Designed a clean, responsive UI with React, implemented user authentication, and live typing indicators with secure data storage in MongoDB.

Online Signature Verification System | *React.js, Flask, Deep Learning (CNN)*

- Developed a **CNN-based Siamese Network** for biometric authentication, achieving **93.07% accuracy** and **0.98 AUC** on the ICDAR 2011 dataset by optimizing Contrastive Loss.
- Built a secure web app using **React.js** and **Flask**, utilizing **MongoDB GridFS** for chunked image storage and implementing **One-Shot Learning** to verify new users without retraining.

Skills

Languages: Java, Python, C

Frontend: React.js, HTML, CSS, TailwindCSS, Bootstrap

Backend: Node.js, Express.js, REST APIs

Databases: MongoDB, SQL

DevOps & Tools: Docker, Git, GitHub, AWS (EC2, ECR)

CS Fundamentals: Data Structures & Algorithms, OOP, OS, DBMS

Achievements

- Published Research Paper:** Authored "Online Signature Verification using Deep Learning" in the **2025 IEEE ICSADL Conference**.
- Selected in the top **1.4%** out of 40,000 applicants for the **AlgoUniversity Accelerator**, receiving mentorship from engineers at Apple & Google.
- Secured **Global Ranks 127, 160, & 190** in CodeChef Starters; **Max Rating: 1612** [CodeChef 
- Achieved **LeetCode Max Rating of 1796** and solved **500+** problems [LeetCode 

Certifications

- Certificate of Excellence (AlgoUniversity):** Awarded for exceptional performance and architectural contributions during the Software Development Externship.
- Certified Smart Coder (Smart Interviews):** Validated expertise in advanced data structures and algorithmic problem-solving 